



MINI PROJECT SYNOPSIS PRESENTATION



AMPHIBIOUS DRONE

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Presentation Overview



- Introduction
- Problem statement
- Literature review
- Aims & objectives
- Proposed System (Methodology)
- Hardware and Software requirements and its needs
- Outcome of the project
- Application and Advantages
- Gantt Chart



Introduction



Unmanned Aerial Vehicles (UAVs) are one of the most revolutionary product lines of modern engineering to come, offering a multitude of applications, from surveillance and disaster management to agriculture and archaeology. Many drones, however, are able to fly only through the air, making the usefulness of these drones limited in conditions with water surfaces. To address this limitation, Amphibious Drone has been proposed.

It is a system that can work well in both the air and the water, an amphibious drone. It possesses the benefits of flying and the ability to land, float and move on water. This thing can do two jobs at once which makes it super useful, for tasks. Like when there is a flood and people need to be rescued or when we want to keep an eye on what is going on in the ocean or even to watch over the coast.



Problem Statement



The problem is to design an amphibious drone that can operate efficiently in both air and water, overcoming limitations of traditional single-environment drone



Literature Review

Table 1 summary of literature review

Authors (Year)	Contributions and Observations
Suresh, R., Jasim, A. A., Nagdev, O. A., Sabin, B. B., Nagaraj, V., & Jeyasimman, D. (2020)	This paper introduces AquaMAV, a novel water jet-propelled jumpglider capable of launching from underwater—a first of its kind. Future developments aim to enable sustained flight and autonomous sensing for improved water data collection.
Esakki, B., Ganesan, S., Mathiyazhagan, S., Ramasubramanian, K., Gnanasekaran, B., Son, B., ... & Choi, J. S. (2018)	An amphibious vehicle combining a multi-rotor UAV and hovercraft was developed for 25-minute missions with a 7 kg payload, featuring a robotic arm and water-suction pump. It was tested in air and water, successfully collecting 500 mL water samples and transmitting IoT-based quality data via 4G-LTE in 11 ms. Sensor analysis showed turbidity and pH saturation times of 65 ms and 26 ms respectively, revealing the lake water was impure per IS 10500 standards.
Saad Bin Abul Kashem, Mujahid Tabassum, Malcolm Chai (2017)	The drone is designed with a high-density, low-volume body to reduce buoyancy and transition time between mediums, enhancing speed and maneuverability. A blended cylindrical shape and quadcopter design minimize drag and power consumption, with simulations and stress tests showing promising results.



Aim & Objectives



AIM:

The aim of this project is to design and develop an amphibious drone capable of efficient operation in both aerial and aquatic environments.

OBJECTIVES:

- ❑ To design a drone that can operate smoothly in both air and water. This will allow for flexible navigation across different environments.
- ❑ To achieve steady movement, control, and transition between flying and underwater modes while losing as little energy as possible.
- ❑ To design propulsion and power systems that reduce energy use while maximizing endurance in both environments.
- ❑ To integrate sensors and cameras for monitoring the environment, surveillance, and collecting data in both air and water.
- ❑ To build a lightweight, strong, and water-resistant body that can handle air resistance and water drag.
- ❑ To create a platform for applications like defence, disaster management, marine exploration, environmental monitoring, and search-and-rescue missions.



Proposed block

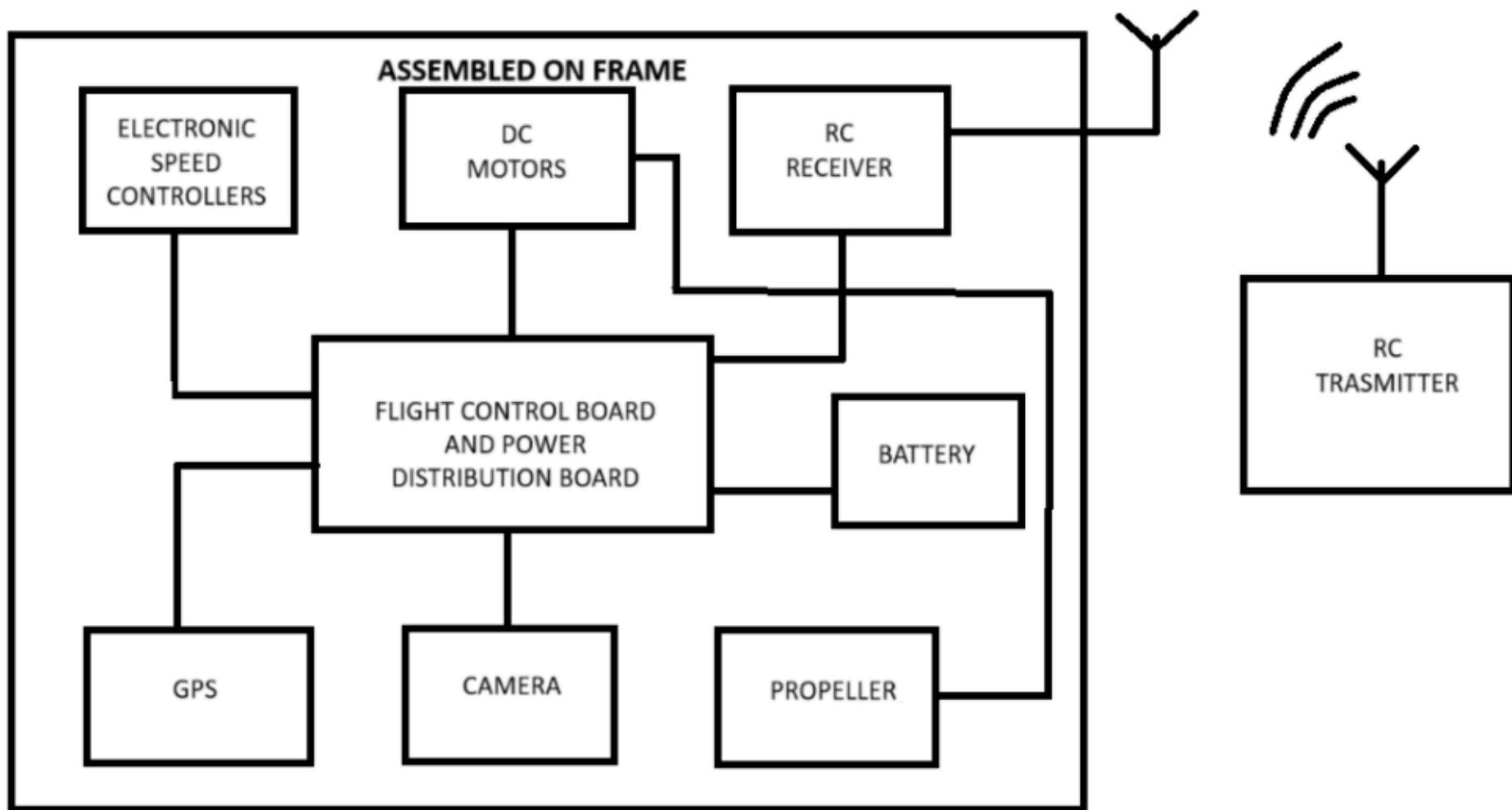


Fig. 1



H/W and S/W Requirements



HARDWARE REQUIREMENTS :

1. Drone Frame
2. Brushless Motor
3. Lipo Battery
4. Electronic Speed Controller
5. Arduino Uno
6. Propeller
7. Transmitter and receiver

SOFTWARE REQUIREMENTS :

1. Arduino ide



Expected outcomes of the project

- Designed a drone that worked well in both air and water, allowing for easy navigation in different environments.
- □ Achieved steady movement, control, and transition between flying and underwater modes while minimizing energy loss.
- Developed propulsion and power systems that cut energy use while maximizing endurance in both settings.
- Integrated sensors and cameras for monitoring, surveillance, and data collection in air and water.
- Built a lightweight, strong, and water-resistant body that withstood air resistance and water drag.
- Created a platform for uses like defence, disaster management, marine exploration, environmental monitoring, and search-and-rescue missions.



Applications and Advantages

APPLICATION:

- ☐ Flood rescue operations
- ☐ Wildlife observation
- ☐ Environmental and industrial monitoring
- ☐ Coastal and boarder surveillance
- ☐ Commercial and recreational use
- ☐ Advanced dual environment surveillance

ADVANTAGES:

- ☐ Dual mode operation
- ☐ Enhanced rescue and safety
- ☐ Cost effective and risk reduction
- ☐ Versatility and efficiency
- ☐ Real time monitoring
- ☐ Infrastructure inspection



Gantt Chart

Activity	Start Week	End Week	Duration
Problem Identification	22/08/2025	28/09/2025	37 days
Literature survey	22/08/2025	01/09/2025	10 days
Circuit/ Block diagram Design	22/08/2025	01/09/2025	10 days
Hardware/Software Development	22/08/2025	10/10/2025	19 days
Testing & Debugging	20/10/2025	25/10/2025	5 days
Documentation & Report Writing	22/08/2025		
Final Presentation Preparation	30/10/2025		



Thank You